

Mahadi Sajjad Nelay

Embedded Systems Engineer

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Professional Experience

08/2023 – present Noakhali, Bangladesh	Embedded Systems Engineer [Onsite] Youth Notion  - Principal Systems Engineer - Project ASTRA - Engineering Coordinator - Project CIPHER - Principal Systems Engineer - Project RAPID - Principal Engineer - Project ARROW - Systems Architect - Project BUTTERFLY - Principal Systems Engineer - Project BLADE
12/2022 – 07/2023 Zhengzhou, Henan, China	Systems Engineer [Remote] Kuayue (Huzhou) Technology Co., Ltd.
07/2021 – 12/2022 Huzhou, Zhejiang, China	Applications Engineer [Remote] Kuayue (Huzhou) Technology Co., Ltd.
01/2020 – 07/2021 Huzhou, Zhejiang, China	Research Analyst [Remote] Cyberspace Security Laboratory - HUTC
09/2017 – 12/2019 Huzhou, Zhejiang, China	Research Assistant [Onsite] IoT Competition Laboratory - HUTC

Education

09/2017 – 07/2021 Huzhou, Zhejiang, China	Bachelor of Engineering School of Information Engineering - Huzhou University  Field of study: Computer Science Thesis: Industrial Automation with ESP8266
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Certificates

OSS Development, Linux and Git Specialization  The Linux Foundation Credential ID : DXLZHCOPYVALL	Google IT Automation Professional Certificate  Google Credential ID : V2VN7MZRSB79
CNSS Certified Network Security Specialist  ICSI (International CyberSecurity Institute), UK Credential ID : 20379392	Deep Learning Specialization  deeplearning.ai Credential ID : XHW5HLZ7888Z
Google IT Support Professional Certificate  Google Credential ID : D5KVGWZSP4UB	Machine Learning Specialization  University of Washington Credential ID : WUHVNWXXQ97WS
Applied Cryptography Specialization  University of Colorado Credential ID : 6CQCH48H23E8	Reinforcement Learning Specialization  Alberta Machine Intelligence Institute (Amii) Credential ID : ULGRB8J8DGHR
SAS Programmer Professional Certificate  SAS Credential ID : MDDULJL48MXE	TensorFlow Developer Professional Certificate  deeplearning.ai Credential ID : 8SUSDYK6JBN2

Languages

Bengali
Native

Mandarin Chinese
HSK 2

English
IELTS Band 8

Skills

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|--------------------|---------|--------|----------|
| • C | • C++ | • Java | • Python |
| • Embedded Systems | • Linux | • Git | • IoT |

Awards

2017	University Scholarship for International Students (Type A) Huzhou University
2018	1st Prize, 3rd Huzhou University IoT Application Innovation Competition Huzhou University
2019	Winner, 2019 National College Student Innovation and Entrepreneurship Training Project People's Republic of China
2020	1st Prize, 5th Huzhou University IoT Application Innovation Competition Huzhou University
2021	Zhejiang Provincial Government Scholarship (Class B) Zhejiang Provincial Government, People's Republic of China

Projects

Acidalia

A real-time face mask detection system developed with python and ported to run on X86 embedded systems running Arch Linux and equipped with 60FPS cameras. It's capable of detecting multiple faces at the same time and flagging the faces of people not wearing a face mask.

This project was specifically requested by the vice president of our faculty, Dr. Honghai, and designed to be installed in public places such as elevators as a measure to prevent Covid.

1211-Alastor

My attempt at engineering an IEEE802.11 and IEEE802.15.1 enabled vehicle that navigates the terrain wirelessly with enhanced mobility, allowing the operator to pilot the platform to perform inspection tasks with the live video stream from the onboard camera and capture data safely, accurately, and frequently regardless of the situation.

Built with an AI Thinker ESP32 CAM, an L298N Dual H-Bridge Motor Controller, an abandoned toy car, and a 12V battery pack.

ARM-Slave

A proof of concept project based on ESP-NOW that allows instantaneous and isolated peer-to-peer communication between multiple nodes in theoretical cases where embedded systems in close proximity need to communicate without a WLAN or internet.

This technique can be used for edge network and edge computing and I intend to use it to deploy a sensor network that feeds sensor data to an AI-powered embedded computer that processes the data in real-time and send feedback to execute tasks without having to rely on a LAN.

References

Dr. Zheng Bo, Professor, School of Information Engineering, HUTC
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Dr. Chen Xiaohua, Associate Professor, School of Information Engineering, HUTC
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Shamir Shehab, Founder & President, Bangladesh Youth Environmental Initiative (BYEI)
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